BRAC University
Department of Mathematics and Natural Sciences

MAT 216: Linear Algebra & Fourier Transformations

Assignment 1

Deadline : 07/03/2025 | Spring 2025 | Total Marks: 50

Make a Front Page by yourself, mentioning your #name, #ID, and #section. (Compulsory)

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1. Discuss the row picture of the following systems, **(5)**

$$\text{(a)}\quad \begin{aligned} a_1x + b_1y &= c_1 \\ a_2x + b_2y &= c_2 \end{aligned} \qquad \text{(b)}\quad \begin{aligned} a_1x + b_1y + c_1z &= d_1 \\ a_2x + b_2y + c_2z &= d_2 \\ a_3x + b_3y + c_3z &= d_3 \end{aligned}$$

2. Solve each of the following system of equations using row echelon form (Gaussian Elimination). Also, comment the type of the solution. $(2 \times 5 = 10)$

$$\text{(a)}\quad \begin{aligned} x_1 + 2x_2 &= -2 \\ 2x_1 + x_2 + x_3 &= 11 \\ 2x_1 + x_3 &= 12 \end{aligned} \qquad \text{(b)}\quad \begin{aligned} 2x_1 + x_2 + 12x_3 &= 11 \\ x_1 + 2x_2 + 9x_3 &= -11 \end{aligned}$$

3. Solve each of the following system of equations using reduced row echelon form (Gauss Jordan Elimination). Also, comment the type of the solution. $(3 \times 5 = 15)$

$$\text{(a)}\quad \begin{aligned} -x_1 + x_2 + x_3 + 2x_4 &= 11 \\ 2x_2 + 2x_3 + x_4 &= 0 \\ 2x_1 + 2x_3 + x_4 &= 12 \end{aligned} \qquad \text{(b)}\quad \begin{aligned} 5x_1 - 5x_2 - 45x_3 + 10x_4 &= 10 \\ -x_1 + x_2 + 9x_3 - 3x_4 &= -11 \end{aligned}$$

$$\text{(c)}\quad -9x_1 - 18x_2 + x_3 - 2x_4 = 9$$

4. For which values of λ, the system of equation has either an unique solution or no solution or infinitely many solution? **(10)**

$$\begin{aligned} x_1 + 2x_2 - x_3 &= 13 \\ 3x_1 - x_2 + 2x_3 &= 31 \\ 4x_1 + x_2 + (\lambda^2 - 9)x_3 &= \lambda + 5 \end{aligned}$$

5. Find the inverse of the following matrices using elementary row operations. Do all the matrices mentioned here are invertible? $(2 \times 5 = 10)$

$$\text{(a)}\ A = \begin{bmatrix} -1 & -6 & -6 \\ 0 & -3 & -4 \\ 2 & 4 & -7 \end{bmatrix} \qquad \text{(b)}\ A = \begin{bmatrix} 3 & -2 & 4 \\ 1 & 0 & -2 \\ 0 & -1 & 0 \end{bmatrix}$$